Note 1.3 Matrix Arithmetic

Matrix Notation

- **Notation:**
  - Use capital letters A, B, C and so on to represent the matrices without writing out all their entries.
  - Use a_{ij} to represent the entry of the matrix A in ith row and jth column and use (i, j) to represent the location of that entry.
  - Use $A = (a_{ij})$ to shortly represent a matrix.

Vector

1. **Scalar:** The entries in the matrices. (Either real or complex numbers)
2. **Vector**:
   - **Vector:** An n-tuple of real numbers that is as a solution of a system of a m linear equations in n unknowns.
     - **Row Vector:** A vector represented by a $1 \times n$ matrix.
       - **notation:** (no universally accepted standard notation)

$$\overrightarrow{\mathbf{x}} = (x_1, x_2, \cdots, x_n)$$

     - **Column Vector:** A vector represented by an $n \times 1$ matrix.(shorten for $vector$ since the almost uses.)
       - $Euclidean\, n\text{-}space$:The set of all vectors of real number. Denoted by $\mathbb{R}^n$.
       - **notation:**

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

     - **Example:** The solution of the linear system $\begin{matrix} x_1 + x_2 = 3 \\ x_1 - x_2 = 1 \end{matrix}$ can be represented by the row vector $(2 \quad 1)$ and the column vector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$.
   - **Notation in matrix**:

- **For jth column:**

$$\mathbf{a}_j = \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix} \qquad j = 1, \cdots, n$$

- **For ith row:** (no universally accepted standard notation)

$$\mathbf{a}_i = (a_{i1}, a_{i2}, \cdots, a_{in}) \qquad i = 1, \cdots, m$$

- **For the matrix *A*:**

$$A = (\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_n) = \begin{pmatrix} \overrightarrow{\mathbf{a_1}} \\ \overrightarrow{\mathbf{a_2}} \\ \vdots \\ \overrightarrow{\mathbf{a_m}} \end{pmatrix}$$

- **Example:** If

$$A = \begin{pmatrix} 3 & 2 & 5 \\ -1 & 8 & 4 \end{pmatrix}$$

then

$$\mathbf{a}_1 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \mathbf{a}_2 = \begin{pmatrix} 2 \\ 8 \end{pmatrix}, \mathbf{a}_3 = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

and

$$\overrightarrow{\mathbf{a}}_1 = (3, 2, 5), \overrightarrow{\mathbf{a}}_2 = (-1, 8, 4)$$

Equality

- **Definition:** Two $m \times n$ matrices A and B are said to be **equal** if $a_{ij} = b_{ij}$ for each i and j.

Scalar Multiplication

- **Definition:** If A is an $m \times n$ matrix and α is a scalar, then αA is the $m \times n$ matrix whose (i, j) entry is αa_{ij}.

Matrix Addition

- **Definition:** If $A = (a_{ij})$ and $B = (a_{ij})$ are both $m \times n$ matrices, then the sum $A + B$ is the $m \times n$ matrix whose (i, j) entry is $a_{ij} + b_{ij}$ for each ordered pair (i, j).

- **Zero matrix:** A matrix whose entries are all zero, denoted by O.
- **Additive inverse**: A matrix adds the matrix A to produce the zero matrix, denoted by $-A$.

Matrix Multiplication and Linear Systems

- **One Equation in Several Unknowns:** For a linear equation with n unknowns of the form

  $$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

  if we let

  $$A = (a_1, a_2, \cdots, a_n) \quad \text{and} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{pmatrix}$$

  and define the product$A\mathbf{x}$ by

  $$A\mathbf{x} = a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

  then the system can be written in the form $A\mathbf{x} = \mathbf{b}$.
- M **Equations in** N **Unknowns:** For a $m \times n$ linear system

  $$\begin{aligned} &a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ &a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ &\vdots \\ &a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{aligned}$$

  then we let

  $$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

  and the product $A\mathbf{x}$ is

  $$A\mathbf{x} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix}$$

then the system can be written in the form $A\mathbf{x}=\mathbf{b}$.It can be understood as

$$A\mathbf{x}=\begin{pmatrix} a_{11}x_1+a_{12}x_2+\cdots+a_{1n}x_n \\ a_{21}x_1+a_{22}x_2+\cdots+a_{2n}x_n \\ \vdots \\ a_{m1}x_1+a_{m2}x_2+\cdots+a_{mn}x_n \end{pmatrix}=\begin{pmatrix} \vec{\mathbf{a_1}}\mathbf{x} \\ \vec{\mathbf{a_2}}\mathbf{x} \\ \vdots \\ \vec{\mathbf{a_n}}\mathbf{x} \end{pmatrix}$$

It can be understood as well as

$$\begin{aligned} A\mathbf{x}&=\begin{pmatrix} a_{11}x_1+a_{12}x_2+\cdots+a_{1n}x_n \\ a_{21}x_1+a_{22}x_2+\cdots+a_{2n}x_n \\ \vdots \\ a_{m1}x_1+a_{m2}x_2+\cdots+a_{mn}x_n \end{pmatrix} \\ &=x_1\begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}+x_2\begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix}+\cdots+x_n\begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} \\ &=x_1\mathbf{a}_1+x_2\mathbf{a}_2+\cdots+x_n\mathbf{a}_n \end{aligned}$$

and the system can be written as $A\mathbf{x}=\mathbf{b}$.

- **Linear combanition:** $c_1, c_2, \cdots, c_n$ are scalars, then a sum of the form

$$c_1\mathbf{a_1}+c_2\mathbf{a_2}+\cdots+c_n\mathbf{a_n}$$

  is said to be a **linear combination** of the vectors $\mathbf{a_1, a_2, \cdots, a_n}$ in $\mathbb{R}^n$.
- **Theorem 1.3.1 Consistency Theorem for linear systems:** A linear system $A\mathbf{x}=\mathbf{b}$ is consistent if and only if $\mathbf{b}$ can be represented as the linear combination of the column vector of A.
- **Matrix Multiplication:**
  - **Conditions:** If and only if A is a $m\times n$ matrix and B is a $n\times q$ matrix they can product a $m\times q$ matrix C.
    - Understanding: The matrix multiplication AB actually is to transform the vector by B and then by A. Matrix B maps the vectors in the q-dimensional space to the n-dimensional space and then matrix A must act on the output of B. So the column of the matrix A must be n to map the vectors in the n-dimensional space to the m-dimensional space. And the transformation equals to $C=AB$, which maps the vector in the q-dimensional space to the m-dimensional space.
  - **Definition:** If $A=(a_{ij})$ is a $m\times n$ matrix and $B=(b_{ij})$ is a $n\times q$ matrix, then the product $AB=C=(c_{ij})$ is a $m\times q$ matrix whose entries are defined by

$$c_{ij}=\vec{\mathbf{a}}_i\mathbf{b}_j=\sum_{k=1}^{n}a_{ik}b_{jk}$$

Actually, it's one row times one column, which is as same as the situation in the situation of upper part.

- **Computation:**
  - By definition. When computing, we can draft the matrix B upper to align the matrix C, like

$$\begin{array}{cc} & \begin{pmatrix} b_{11} & \cdots & b_{1q} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{nq} \end{pmatrix} \\ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} & \begin{pmatrix} c_{11} & \cdots & c_{1q} \\ \vdots & & \vdots \\ c_{m1} & \cdots & c_{mq} \end{pmatrix} \end{array}$$

  - **By the linear combination of the row vector:** The product AB can be understood by

$$AB = \begin{pmatrix} \vec{\mathbf{a}}_1 B \\ \vec{\mathbf{a}}_2 B \\ \vdots \\ \vec{\mathbf{a}}_n B \end{pmatrix}$$

    then we can understand the rows in the C as the linear combination of the row vectors of B.(See also Partition into rows)
  - **By the linear combination of the column vector:** The product AB can be understood by

$$AB = (A\mathbf{b}_1, A\mathbf{b}_2, \cdots, A\mathbf{b}_n)$$

    then we can use the same method of the upper part, which understands the columns in C as the linear combination of the column vectors of A.(See also Partition into columns)
  - **By the sum of the columns of A times the rows of B:** The outcome of one column times one row is a matrix. So the product AB can be understood by

$$AB = \mathbf{a}_1\vec{\mathbf{b}}_1 + \mathbf{a}_2\vec{\mathbf{b}}_2 + \cdots + \mathbf{a}_n\vec{\mathbf{b}}_n$$

for example,

$$\begin{pmatrix} 3 & -2 \\ 2 & 4 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} -2 & 1 & 3 \\ 4 & 1 & 6 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} \begin{pmatrix} -2 & 1 & 3 \end{pmatrix} + \begin{pmatrix} -2 \\ 4 \\ -3 \end{pmatrix} \begin{pmatrix} 4 & 1 & 6 \end{pmatrix}$$
$$= \begin{pmatrix} -6 & 3 & 9 \\ -4 & 2 & 6 \\ -2 & 1 & 3 \end{pmatrix} + \begin{pmatrix} -8 & -2 & -12 \\ 16 & 4 & 24 \\ -12 & -3 & -18 \end{pmatrix}$$
$$= \begin{pmatrix} -14 & 1 & -3 \\ 12 & 6 & 30 \\ -14 & -2 & -15 \end{pmatrix}$$

(See also Outer product expansion)

- **Remarks:** $AB \neq BA$.

Notational Rules

- Follow the rules of arithmetic, accosiative law and distributive law(no commutative law).

The Transpose of a Matrix

- **Definition:** The **transpose** of an $m \times n$ matrix A is the $n \times m$ matrix B defined by

$$b_{ji} = a_{ij}$$

  for $j = 1, \cdots, n$ and $i = 1, \cdots, m$. The transpose of A is denoted by A^T. The diagonal entries of A and A^T are the same.

- **Example:** If $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$, then $A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$.
- **Symmetric matrix:** An $n \times n$ matrix A is said to be symmetric if $A^T = A$.

Application 1 Production Costs

- **Strategy:** Make two matrix of amount and expenses and multiple them to get a new matrix of the total costs.
- **Example:** The example of production costs

Application 2 Information Retrieval

- **Strategy:** Suppose a database consisting n documents and there are m keywords, we can represent the database by an $m \times n$ matrix A. Then we take a_{ij} as the frequency of the jth keyword appearing in the ith document.Then the searching keywords are represented by a vector $\mathbf{x}$ in $\mathbb{R}^m$. Take the 1 and 0 to represent whether the keyboard appears in the list and simply multiply A^T times $\mathbf{x}$.

- **Example:**
  - Simple Matching Searches
  - Relative-Frequency Searches

Section 1.3 Exercises

- Question 11th

  For given information, we can get

  $$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

  so the system is consistent. Since there are two given solution, A must involve a free variable. Thus the system has infinitely many solutions.
- Question 12th

  For given information, we can get

  $$\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

  so the system is consistent. Since A is a 3×4 matrix, there are most 3 lead variables. Thus there exists at least one free variable in A. Then the system has infinitely many solutions.
- Question 13th
  - Since the augmented matrix has reduced row echelon form and x_2, x_4, x_5 are free variables, then we let $x_2 = a, x_3 = b, x_5 = c$. Hence

    $$\mathbf{x} = (-2 - 2a - 3b - c, a, 5 - 2b - 5c, b, c)^T$$

    is the form that the solution consists of all vectors with this form.
  - No matter what the free variables are, since $\mathbf{a}_1$and $\mathbf{a}_3$ are given, the corresponding $\mathbf{b}$ is unique. Let x_2, x_4, x_5 be zero, then $\mathbf{x}_0 = (-2, 0, 5, 0, 0)^T$ is a solution to $A\mathbf{x} = \mathbf{b}$ and hence

    $$\mathbf{b} = A\mathbf{x}_0 = -2\mathbf{a}_1 + 5\mathbf{a}_3 = (8, -7, -1, 7)^T$$

- Question 15th

  For A^T, the diagonal entries is a_{jj} at (j, j). For $-A$, the diagonal entries is $-a_{jj}$ at (j, j). Since $A^T = -A$, then $-a_{jj} = a_{jj}$, thus $a_{jj} = 0$. So the diagonal entries must all be 0.

Definition:
A matrix A is skew symmetric if $A^T = -A$.